



# TOTALITY

A generator of hypnotic Berlin techno

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## About

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Totality composes, mixes and plays hypnotic Berlin techno: tracks of six to eight minutes and DJ sets of up to four hours, in which the tracks follow one another on two decks as a DJ would mix them. Nothing is played back from a recording. The kick, the rumble under it, the twelve lanes of the drum kit, the ping, the bass and the 303, the dub chord, the drone and the texture are synthesised; the tape echo, the spring, the plate and the rooms are simulated; and every note, every operation of the form and every move of a knob is decided by the composer.

The music it aims at is the one of Berghain and Ostgut Ton, of Dub Techno and of the hypnotic school: a groove that holds for minutes and changes by one element at a time, a low end made of a kick and the rumble of its own reverb, a chord thrown into a tape echo. Four style profiles span it -- Hypnotic in the centre, Ostgut, Dub and Raw -- and every knob of their recipes was fitted against measurements of thirty reference tracks: their spectral balance, their width, their loudness and its range, and how alike their bars are.

The same seed always gives the same track. A track is a seed, a style, a length and a handful of rerolls, which is what a set file stores and what you can hand to someone else. The standalone and the plugin play what the offline renderer plays, sample for sample.

Totality is the moment of a total eclipse when the moon covers the sun entirely and only the corona shines round the dark disc. In Totality's Patterns view the kick is that disc: every other part turns around it in rings, and its corona swells with every beat.

## Quick start

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The standalone plays by itself: press Play. Compose writes a new track with the knobs as they stand, New seed draws a fresh one. The status line tells what plays: its style and form, its tempo, its key with the Camelot label a DJ mixes by, and its length. While a track begins to play, Totality measures its loudest part in the background and brings it to its style's loudness; the status says "levelling" until the correction has glided in.

For a set, move Set min above zero: Totality then writes a set of that many minutes -- a dramaturgy of tempo and energy, tracks in neighbouring keys, blends of 16 or 32 bars with the bass swapped on the 32-bar line, loops of the last track under the next on the third deck, breaks from the mixer's effects.

In a DAW, Totality is an instrument plugin that follows the host's transport: it plays when the host plays, jumps when the host jumps, and plays at the host's tempo (a set's own tempo drift is the standalone's and the export's). Its output is a finished stereo mix at about -10 LUFS with its true peak under -1 dBTP.

A reroll draws one part of the track again and leaves the rest as it was: form, harmony, rack, layers, blocks, events, hands, sounds. In a set the rerolls act on the track under the playhead, and "reroll the whole track" or "the set's plan" draw more at once (the Arrange page).

# The track and the set



A track has one of three forms. The Arc is the DJ tool: an intro of 32 or 64 bars from the kick on, a body that adds one layer every 32-bar block, an outro that takes them away again -- so it mixes. The Peak builds to a long kick-out and returns into its densest block. The Endless is full from the first bar and changes only by exchanging layers, as the hypnotic records do. Every block has exactly one operation (add, remove, swap) on its first bar -- a hat or a perc may come in on the block's 8- or 16-bar line instead --, every change falls on a 4-bar line and a return on a 16-bar line; nothing tonal plays in the first and last 32 bars, so two tracks can be mixed without a clash.

Every track has a figure: the voice it is remembered by -- a ping motif for Hypnotic, a chord stab or a bass riff for Ostgut, the dub chord for Dub, a 303 line for Raw. It plays the same one or two bars from its entry (the body's second block; a bass riff with the body) to the end and changes only through its filters. Between the operations the body runs in waves of 32 or 64 bars: the figure's filters and the hats' bus open towards a wave's end and fall back after it; two to four bars before that landing something drops away -- the centre of the mix, the kick and the claps, the figure -- so that the landing lands; in the middle of a wave one element (the figure, the bass or the rumble, the percussion, the open hats and the ride) breathes out for 8 or 16 bars and comes back with an echo throw. The Arrange page names the figure and the bars where the waves land.

Style picks the profile; Morph To and Morph blend it with a second one; the Dub and Hypnotic shares pull any profile along the two axes the style is described by. With Auto the tempo, the key and the owner of the low end are drawn from the profile; without it the knobs decide. Minutes is the track's length, rounded to whole blocks.

The set knobs: its length, the dramaturgy (Warm-up, Peak, Closing, Sunday, Flat -- the arc of tempo and energy over its length), whether the styles wander with the energy or stay, how often a loop of the outgoing

track plays on under the next one, how often a break comes from the mixer's effects, and the blend's length. The mixer's effects are one send per deck into a tape echo and a long hall, which the set's breaks throw into and the Perform page's throw uses.

## COMPOSE

| Control           | Key                                 | Range                                                  | Default  |
|-------------------|-------------------------------------|--------------------------------------------------------|----------|
| Tempo             | <code>compose.bpm</code>            | 100 .. 160 BPM                                         | 130 BPM  |
| Key               | <code>compose.key</code>            | C, C#, D, D#, E, F, F#, G, G#, A, A#, B                | A        |
| Scale             | <code>compose.scale</code>          | Aeolian, Dorian, Phrygian, Hexachord, Minor Pentatonic | Aeolian  |
| Style             | <code>compose.style</code>          | Hypnotic, Ostgut, Dub, Raw Peak                        | Hypnotic |
| Length            | <code>compose.minutes</code>        | 2 .. 16 min                                            | 7 min    |
| Swing             | <code>compose.swing</code>          | 50 .. 66 %                                             | 53 %     |
| Humanize          | <code>compose.humanize</code>       | 0 .. 10 ms                                             | 3 ms     |
| Low End           | <code>compose.low_owner</code>      | Rumble, Sub                                            | Rumble   |
| Auto              | <code>compose.auto</code>           | off / on                                               | on       |
| Form              | <code>compose.form</code>           | Auto, Arc, Peak, Endless                               | Auto     |
| Morph To          | <code>compose.morph_to</code>       | Off, Hypnotic, Ostgut, Dub, Raw Peak                   | Off      |
| Morph             | <code>compose.morph</code>          | 0 .. 1                                                 | 0.5      |
| Dub Share         | <code>compose.dub_share</code>      | 0 .. 1                                                 | 0        |
| Hypnotic Share    | <code>compose.hypnotic_share</code> | 0 .. 1                                                 | 0        |
| Composer's Sounds | <code>compose.pick_sounds</code>    | off / on                                               | on       |

## SET

| Control    | Key                         | Range                                | Default |
|------------|-----------------------------|--------------------------------------|---------|
| Dramaturgy | <code>set.dramaturgy</code> | Warm-up, Peak, Closing, Sunday, Flat | Peak    |
| Styles     | <code>set.journey</code>    | Stay, Wander                         | Wander  |
| Live Loops | <code>set.loops</code>      | 0 .. 1                               | 0.3     |
| FX Breaks  | <code>set.fx_breaks</code>  | 0 .. 1                               | 0.3     |
| Blend      | <code>set.blend</code>      | 16 bars, 32 bars                     | 32 bars |
| Set Length | <code>set.minutes</code>    | 0 .. 240 min                         | 0 min   |

## DJFX

| Control     | Key                           | Range                          | Default |
|-------------|-------------------------------|--------------------------------|---------|
| Echo Time   | <code>djfx.echo_time</code>   | 1/16, 1/8, 3/16, 1/4, 3/8, 1/2 | 3/16    |
| Feedback    | <code>djfx.feedback</code>    | 0 .. 1                         | 0.45    |
| Echo Return | <code>djfx.echo_return</code> | -60 .. 6 dB                    | -3 dB   |
| Hall Decay  | <code>djfx.hall_decay</code>  | 0.5 .. 12 s                    | 5 s     |
| Hall Return | <code>djfx.hall_return</code> | -60 .. 12 dB                   | 6 dB    |

## Sounds and presets

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Every synth has 1024 factory presets: the kick, the rumble, the sub, a lane of the kit, the ping, the bass, the 303, the dub chord, the drone and the texture. Each engine has sixteen groups -- "Berlin 909", "Dub Sine", "Ride Bell", "Classic 303", "Basic Channel" -- and each group sixty-four presets on an eight by eight grid: the rows an adjective from dark to bright ("Muffled" ... "Cutting"), the columns a noun of the group's own, so the name tells where on the grid a sound lies. A preset is the sound, not the mix: levels, pans, sends, ducking, the octaves and a lane's role and pattern stay where the mix and the composer put them.

For every track the composer chooses a preset per synth and per lane of the kit (Composer's Sounds, `compose.pick_sounds`): a group by how well it suits the track's style, a lane only among the presets made for its role, and in the group a row that leans to the middle -- the darkest and the brightest are chosen less often. The style then holds a few knobs in its own range (the Dub kick's soft click, the Raw kick's hard clip, the Dub chord's bright band). Reroll sounds draws other presets and leaves the notes as they are.

The knobs show what plays. When a track begins -- in a set, when the next track starts on its deck -- its presets and its style's mix are put on the knobs, and each page names the preset the composer chose ("this track: Punchy Drum Hall (Hypnotic Thud)"); the Arrange page lists them for the whole track. Turn a knob and the sound follows from there; choose another preset from the menu, or step through them, and it plays at once. In a blend each deck keeps its own track's sound: the knobs show the incoming track while the outgoing one plays on as it was.

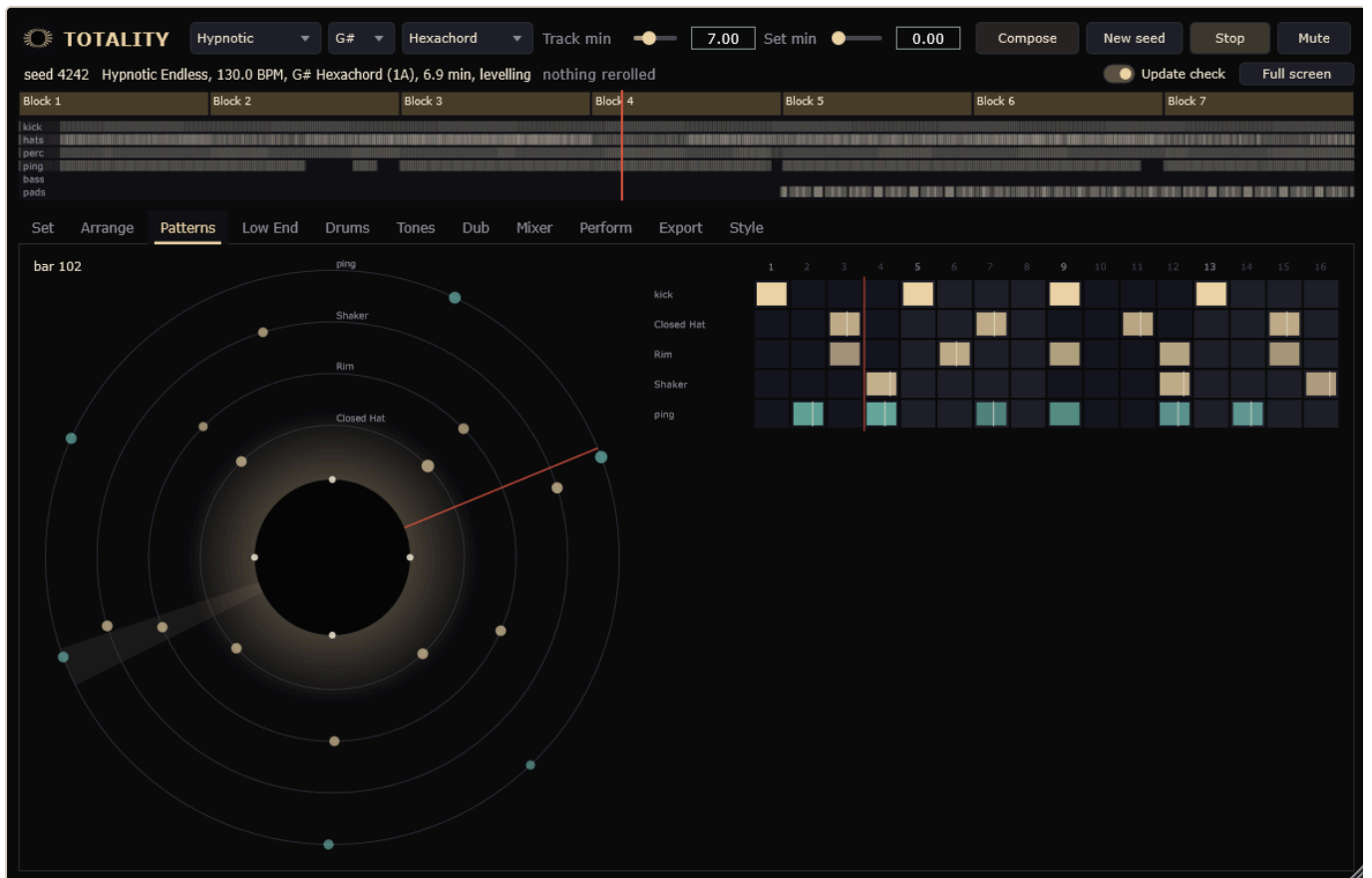
# Arrange

The screenshot displays the TOTALITY software interface, specifically the 'Arrange' page. At the top, the track name 'Hypnotic Endless' is shown along with its tempo (130.0 BPM) and key (G# Hexachord (1A)). The interface is divided into several sections. The top section shows a timeline with seven blocks (Block 1 to Block 7) and a playhead at Block 4. Below this, there are tabs for 'Set', 'Arrange', 'Patterns', 'Low End', 'Drums', 'Tones', 'Dub', 'Mixer', 'Perform', 'Export', and 'Style'. The 'Arrange' tab is active, showing a detailed view of the track's structure. A text box at the top of the Arrange page provides information about the track: 'The track: Hypnotic, Endless, 130.0 BPM, G# Hexachord (1A), 224 bars, the rumble owns the low end, bar similarity 0.78; figure: the ping from bar 1, waves land on 65 129 193'. Below this, a list of sounds is provided: 'kick Punchy Drum Hall, rumble Noon Transept, sub Round Spell, ping Golden Reservoir, bass Round Wheel, acid Waxy Patch, chord Floating Gibbous, drone Warm Tarnish, texture Aged Cable'. A row of buttons allows for various operations: 'reroll form', 'reroll harmony', 'reroll rack', 'reroll layers', 'reroll blocks', 'reroll events', 'reroll hands', 'reroll sounds', and 'reroll figure'. The main area of the Arrange page shows a grid of tracks (kick, hats, perc, ping, bass, pads) across the seven blocks. A vertical red line indicates the current position of the playhead at Block 4. The bottom section shows a detailed view of the tracks, with a 'start kick' button at the beginning of Block 1 and an 'add ride' button at the end of Block 2. The tracks are labeled 'kick', 'hats', 'perc', 'ping', 'bass', and 'pads'.

The arrangement across the whole length: in a set the tracks on decks A and B (their swaps marked: from there a track owns the low end), in a track its blocks with their names, and under them a lane per group of layers -- kick, hats, perc, ping, bass, pads -- lit where it plays. The Arrange page shows the operations of the form by name. A click jumps there.

The row of rerolls acts on the track under the playhead; each part of it is drawn from a stream of its own, so a reroll of the blocks keeps the harmony and the sounds, a reroll of the hands (the automation) keeps every note, and a reroll of the figure draws another motif for the same voice.

# Patterns and the Eclipse



The kick is the dark disc in the middle, and every other part that plays in the bar is a ring around it: a bar once round, its onsets as beads, larger the louder, the playhead's hand sweeping over them. A layer on the sixteen-step matrix shows the same beads bar after bar; a polymeeter or a slipping layer moves on from bar to bar -- it precesses, and you can see it. Where three rings or more have an onset on the same sixteenth, the corona lights up at that angle: a conjunction. Beside it the same bar as a step grid, with a tick where an onset lies off the grid (swing and feel).

Hypnotic

G#

Hexachord

Track min

7.00

Set min

0.00

Compose

New seed

Stop

Mute

seed 4242 Hypnotic Endless, 130.0 BPM, G# Hexachord (1A), 6.9 min, levelling nothing rerolled

Block 1

Block 2

Block 3

Block 4

Block 5

Block 6

Block 7

kick

hats

perc

ping

bass

pads

Set

Arrange

Patterns

Low End

Drums

Tones

Dub

Mixer

Perform

Export

Style

KICK PRESET

<

Punchy Drum Hall

>

this track: Punchy Drum Hall (Hypnotic Thud)

ENGINE

Engine

Tune

Level

Drive

Clip

909

Key

-3.00 dB

0.29

Soft

PITCH

Start

End

Body Decay

261 Hz

48.7 Hz

23.8 ms

PUNCH AND AMP

Punch

Punch Decay

Attack

Hold

Decay

Tail Limit

0.48

3.00 ms

0.20 ms

25.0 ms

450 ms

0.00 dB

CLICK

Click

Tone

Decay

0.22

3000 Hz

6.00 ms

TONE

Tone

Low Cut

Dip Freq

Dip

4245 Hz

30.0 Hz

500 Hz

-3.97 dB

TOP LAYER

Top Level

Top Pitch

Top Decay

Top Drive

Top Cut

-16.1 dB

-8.00 st

90.0 ms

0.60

400 Hz

RUMBLE PRESET

<

Noon Transept

>

this track: Noon Transept (Cathedral Low)

RUMBLE

Level

Split

Drive

Resonance

-9.00 dB

71.3 Hz

4.69 dB

0.15

ROOM

Decay

Size

Damping

Pre-Delay

Low Pass

2.60 s

2.10

0.54

24.7 ms

3.00 x f0

SUB

Sub

Attack

Release

0.00 dB

25.0 ms

300 ms

DUCK

Duck

Hold

Release

7.00 dB

80.0 ms

220 ms

SUB PRESET

<

Round Spell

>

this track: Round Spell (Hypnotic Sub)

SUB

Level

Octave

Drive

Low Pass

Kick Lock

-6.00 dB

0

0.12

129 Hz

Kick

ENVELOPE

Attack

Decay

Sustain

Release

3.35 ms

282 ms

0.71

112 ms

DUCK

Duck

Hold

Release

9.00 dB

40.0 ms

180 ms

has three engines (a swept sine, a resonator, a modelled 909) and a top layer; its drive runs at four times the rate.

The rumble is the kick sent into a hall, cut below the split, clipped and ducked by the next kick -- the dark roll of Berlin techno. Either the rumble or the sub bass owns the band under 80 Hz (compose.low\_owner): never both at once.

## KICK

| Control     | Key              | Range                          | Default |
|-------------|------------------|--------------------------------|---------|
| Engine      | kick.engine      | Sweep, Resonator, 909          | Sweep   |
| Tune        | kick.tune        | Free, Key, Fifth, Flat Seventh | Key     |
| Pitch End   | kick.pitch_end   | 30 .. 120 Hz                   | 52 Hz   |
| Pitch Start | kick.pitch_start | 60 .. 1500 Hz                  | 300 Hz  |
| Body Decay  | kick.pitch_decay | 5 .. 300 ms                    | 18 ms   |
| Punch Decay | kick.punch_decay | 0.5 .. 20 ms                   | 3 ms    |
| Punch       | kick.punch       | 0 .. 1                         | 0.45    |
| Attack      | kick.amp_attack  | 0 .. 10 ms                     | 0.2 ms  |
| Hold        | kick.amp_hold    | 0 .. 150 ms                    | 25 ms   |
| Decay       | kick.amp_decay   | 20 .. 1500 ms                  | 380 ms  |
| Drive       | kick.drive       | 0 .. 1                         | 0.35    |
| Clip        | kick.clip        | Soft, Hard                     | Soft    |
| Click       | kick.click_level | 0 .. 1                         | 0.35    |
| Click Tone  | kick.click_tone  | 500 .. 12000 Hz                | 3000 Hz |
| Click Decay | kick.click_decay | 0.5 .. 30 ms                   | 6 ms    |
| Tone        | kick.tone        | 200 .. 20000 Hz                | 6000 Hz |
| Level       | kick.level       | -36 .. 6 dB                    | -3 dB   |
| Tail Limit  | kick.tail_limit  | -60 .. 0 dB                    | 0 dB    |
| Low Cut     | kick.low_cut     | 15 .. 60 Hz                    | 30 Hz   |
| Dip Freq    | kick.dip_freq    | 200 .. 1500 Hz                 | 500 Hz  |
| Dip         | kick.dip         | -12 .. 0 dB                    | -3 dB   |
| Top Level   | kick.top_level   | -60 .. 0 dB                    | -12 dB  |
| Top Pitch   | kick.top_pitch   | -24 .. 0 st                    | -8 st   |
| Top Decay   | kick.top_decay   | 20 .. 400 ms                   | 90 ms   |
| Top Drive   | kick.top_drive   | 0 .. 1                         | 0.6     |
| Top Cut     | kick.top_cut     | 150 .. 2000 Hz                 | 400 Hz  |

## RUMBLE

| Control | Key          | Range        | Default |
|---------|--------------|--------------|---------|
| Level   | rumble.level | -60 .. 6 dB  | -9 dB   |
| Split   | rumble.split | 50 .. 150 Hz | 80 Hz   |
| Decay   | rumble.decay | 0.5 .. 6 s   | 1.8 s   |

|              |                     |               |        |
|--------------|---------------------|---------------|--------|
| Size         | rumble.size         | 0.3 .. 3      | 1      |
| Damping      | rumble.damping      | 0 .. 1        | 0.5    |
| Pre-Delay    | rumble.predelay     | 0 .. 60 ms    | 0 ms   |
| Drive        | rumble.drive        | 0 .. 18 dB    | 6 dB   |
| Low Pass     | rumble.ratio        | 1.5 .. 6 x f0 | 3 x f0 |
| Resonance    | rumble.resonance    | 0 .. 1        | 0.15   |
| Sub          | rumble.sub          | -60 .. 6 dB   | 0 dB   |
| Sub Attack   | rumble.sub_attack   | 5 .. 200 ms   | 25 ms  |
| Sub Release  | rumble.sub_release  | 50 .. 2000 ms | 300 ms |
| Duck         | rumble.duck         | 0 .. 24 dB    | 7 dB   |
| Duck Hold    | rumble.duck_hold    | 0 .. 250 ms   | 80 ms  |
| Duck Release | rumble.duck_release | 30 .. 600 ms  | 220 ms |

## SUB

| Control      | Key              | Range         | Default |
|--------------|------------------|---------------|---------|
| Level        | sub.level        | -60 .. 6 dB   | -6 dB   |
| Octave       | sub.octave       | -2 .. 2       | 0       |
| Attack       | sub.attack       | 0.5 .. 50 ms  | 3 ms    |
| Decay        | sub.decay        | 20 .. 2000 ms | 250 ms  |
| Sustain      | sub.sustain      | 0 .. 1        | 0.7     |
| Release      | sub.release      | 10 .. 800 ms  | 90 ms   |
| Low Pass     | sub.low_pass     | 40 .. 300 Hz  | 110 Hz  |
| Drive        | sub.drive        | 0 .. 1        | 0.1     |
| Kick Lock    | sub.lock         | Off, Kick     | Kick    |
| Duck         | sub.duck         | 0 .. 24 dB    | 9 dB    |
| Duck Hold    | sub.duck_hold    | 0 .. 250 ms   | 40 ms   |
| Duck Release | sub.duck_release | 30 .. 600 ms  | 180 ms  |

# Drums



Twelve lanes, each a small drum synthesiser: a role (the closed and rolling hats, the open hat, the ride, the claps, the snare, the rim, the shaker, toms and congas, noise), an engine (noise, metal -- the 909's six square waves -- modal, tone, FM), a filter, a shape. They run side by side in the lanes of the processor's vector registers. The hats and the percussion each have a bus with its level and a low pass, then the drum bus's saturation; the ducks let the pads and the rooms step aside for the kick, band by band.

## PERC (12 INSTANCES)

| Control      | Key                | Range                                                                                            | Default    |
|--------------|--------------------|--------------------------------------------------------------------------------------------------|------------|
| Active       | percN.active       | off / on                                                                                         | on         |
| Role         | percN.role         | Closed Hat, Rolling Hat, Open Hat, Ride, Clap, Clap Ghost, Snare, Rim, Shaker, Tom, Conga, Noise | Closed Hat |
| Engine       | percN.engine       | Noise, Metal, Modal, Tone, FM                                                                    | Noise      |
| Pitch        | percN.pitch        | 40 .. 12000 Hz                                                                                   | 400 Hz     |
| Pitch Amount | percN.pitch_amount | 1 .. 16 x                                                                                        | 1 x        |
| Pitch Decay  | percN.pitch_decay  | 0.5 .. 300 ms                                                                                    | 10 ms      |
| FM Ratio     | percN.fm_ratio     | 0.25 .. 8                                                                                        | 1.41       |
| FM Index     | percN.fm_index     | 0 .. 8                                                                                           | 0          |
| Modes        | percN.mode_set     | Membrane, Bar, Harmonic                                                                          | Membrane   |

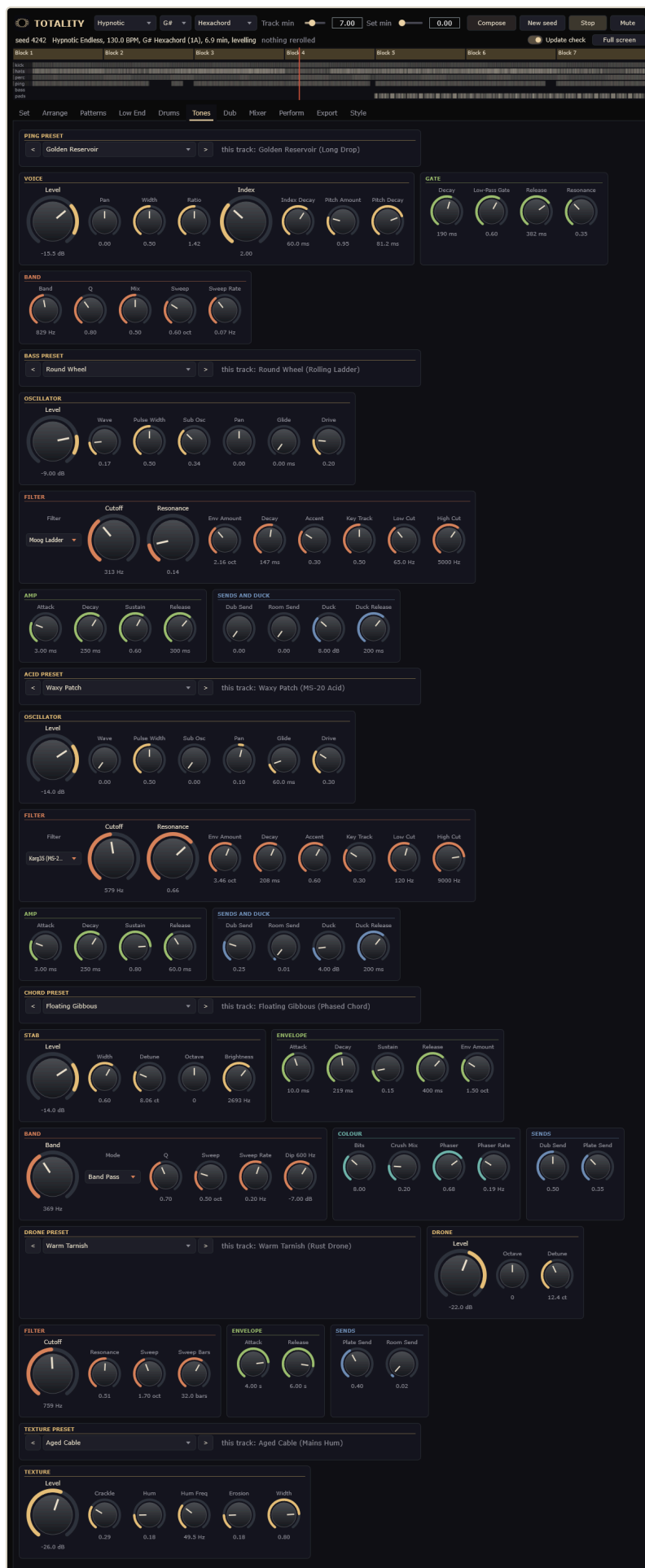
|                  |                     |                                |             |
|------------------|---------------------|--------------------------------|-------------|
| Mode Damping     | percN.mode_damp     | 0 .. 1                         | 0.5         |
| Metal Scale      | percN.metal_scale   | 0.25 .. 4 x                    | 1 x         |
| Noise            | percN.noise         | 0 .. 1                         | 1           |
| Noise Decay      | percN.noise_decay   | 2 .. 3000 ms                   | 70 ms       |
| Bursts           | percN.bursts        | 1 .. 6                         | 1           |
| Burst Spacing    | percN.burst_spacing | 2 .. 40 ms                     | 10 ms       |
| Decay            | percN.decay         | 2 .. 3000 ms                   | 70 ms       |
| Filter           | percN.filter        | Low Pass, Band Pass, High Pass | High Pass   |
| Cutoff           | percN.cutoff        | 100 .. 18000 Hz                | 7000 Hz     |
| Resonance        | percN.resonance     | 0 .. 1                         | 0.15        |
| Low Cut          | percN.low_cut       | 150 .. 8000 Hz                 | 3000 Hz     |
| Drive            | percN.drive         | 0 .. 1                         | 0.1         |
| Level            | percN.level         | -36 .. 6 dB                    | -0.7 dB     |
| Pan              | percN.pan           | -1 .. 1                        | 0.1         |
| Choke Group      | percN.choke         | 0 .. 4                         | 1           |
| Shift            | percN.shift         | -10 .. 10 ms                   | 0 ms        |
| Density          | percN.density       | 0 .. 1                         | 0.5         |
| Tune to Key      | percN.tune          | off / on                       | off         |
| Pan Depth        | percN.pan_depth     | 0 .. 1                         | 0.3         |
| Pan Period       | percN.panBars       | 0.0625 .. 16 bars              | 0.1875 bars |
| Cut Tracks Pitch | percN.cut_track     | 0 .. 2                         | 0           |
| Noise Type       | percN.noise_type    | White, 909 Metal               | 909 Metal   |

## MIX

| Control         | Key            | Range           | Default  |
|-----------------|----------------|-----------------|----------|
| Hats Level      | mix.hats_level | -24 .. 12 dB    | -3.5 dB  |
| Hats Cut        | mix.hats_cut   | 200 .. 20000 Hz | 20000 Hz |
| Perc Level      | mix.perc_level | -24 .. 12 dB    | -6.5 dB  |
| Perc Cut        | mix.perc_cut   | 200 .. 20000 Hz | 20000 Hz |
| Drum Saturation | mix.drum_sat   | 0 .. 1          | 0.35     |
| Duck Low        | mix.duck_low   | 0 .. 18 dB      | 10 dB    |
| Duck Mid        | mix.duck_mid   | 0 .. 12 dB      | 3 dB     |
| Group Low Cut   | mix.low_cut    | 20 .. 500 Hz    | 20 Hz    |

# Tones

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The ping is the hypnotic school's figure: FM through a low-pass gate, a band pass that wanders. The bass and the 303 share one synthesiser with ten circuit filters (the Moog ladder, the 303's diode ladder, the Prophet's SSM2040, the Juno's IR3109, the SEM, the MS-20's Korg35 and more); the 303 line has its accents and slides. The dub chord is a minor seventh or a stab of six voices through a band pass, a crusher and a phaser, sent into the tape echo and the plate. The drone holds a note under eight bars; the texture is vinyl crackle, hum and erosion.

## PING

| Control       | Key               | Range          | Default  |
|---------------|-------------------|----------------|----------|
| Level         | ping.level        | -60 .. 6 dB    | -12.5 dB |
| Pan           | ping.pan          | -1 .. 1        | 0        |
| Width         | ping.width        | 0 .. 1         | 0.5      |
| Ratio         | ping.ratio        | 0.25 .. 8      | 1.414    |
| Index         | ping.index        | 0 .. 6         | 2        |
| Index Decay   | ping.index_decay  | 2 .. 500 ms    | 60 ms    |
| Pitch Amount  | ping.pitch_amount | 0 .. 4         | 0.8      |
| Pitch Decay   | ping.pitch_decay  | 2 .. 300 ms    | 45 ms    |
| Decay         | ping.decay        | 10 .. 2000 ms  | 140 ms   |
| Low-Pass Gate | ping.lpg          | 0 .. 1         | 0.6      |
| Gate Release  | ping.lpg_release  | 10 .. 2000 ms  | 150 ms   |
| Resonance     | ping.resonance    | 0 .. 1         | 0.35     |
| Band          | ping.band         | 150 .. 6000 Hz | 900 Hz   |
| Band Q        | ping.band_q       | 0.2 .. 8       | 0.8      |
| Band Mix      | ping.band_mix     | 0 .. 1         | 0.5      |
| Sweep         | ping.sweep        | 0 .. 2 oct     | 0.6 oct  |
| Sweep Rate    | ping.sweep_rate   | 0.01 .. 2 Hz   | 0.07 Hz  |

## BASS

| Control     | Key              | Range                                                                                                                              | Default       |
|-------------|------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Level       | bass.level       | -60 .. 6 dB                                                                                                                        | -9 dB         |
| Pan         | bass.pan         | -1 .. 1                                                                                                                            | 0             |
| Wave        | bass.wave        | 0 .. 1                                                                                                                             | 0             |
| Pulse Width | bass.pulse_width | 0.05 .. 0.95                                                                                                                       | 0.5           |
| Sub Osc     | bass.sub_osc     | 0 .. 1                                                                                                                             | 0.3           |
| Filter      | bass.filter      | Moog Ladder, Prophet (SSM2040), Juno (IR3109), Oberheim SEM, Xpander, Diode Ladder (303), Korg35 (MS-20), Polivoks, EDP Wasp, Comb | Juno (IR3109) |
| Cutoff      | bass.cutoff      | 40 .. 12000 Hz                                                                                                                     | 350 Hz        |
| Resonance   | bass.resonance   | 0 .. 1                                                                                                                             | 0.2           |
| Env Amount  | bass.env_amount  | 0 .. 6 oct                                                                                                                         | 2 oct         |
| Decay       | bass.decay       | 5 .. 3000 ms                                                                                                                       | 180 ms        |

|              |                   |                 |         |
|--------------|-------------------|-----------------|---------|
| Accent       | bass.accent       | 0 .. 1          | 0.3     |
| Attack       | bass.amp_attack   | 0.5 .. 500 ms   | 3 ms    |
| Amp Decay    | bass.amp_decay    | 5 .. 3000 ms    | 250 ms  |
| Sustain      | bass.amp_sustain  | 0 .. 1          | 0.6     |
| Release      | bass.amp_release  | 5 .. 3000 ms    | 300 ms  |
| Glide        | bass.glide        | 0 .. 500 ms     | 0 ms    |
| Drive        | bass.drive        | 0 .. 1          | 0.2     |
| Key Track    | bass.key_track    | 0 .. 1          | 0.5     |
| Low Cut      | bass.low_cut      | 20 .. 500 Hz    | 65 Hz   |
| High Cut     | bass.high_cut     | 500 .. 20000 Hz | 5000 Hz |
| Dub Send     | bass.dub_send     | 0 .. 1          | 0       |
| Room Send    | bass.room_send    | 0 .. 1          | 0       |
| Duck         | bass.duck         | 0 .. 24 dB      | 8 dB    |
| Duck Release | bass.duck_release | 30 .. 600 ms    | 200 ms  |

## ACID

| Control     | Key              | Range                                                                                                                              | Default            |
|-------------|------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Level       | acid.level       | -60 .. 6 dB                                                                                                                        | -14 dB             |
| Pan         | acid.pan         | -1 .. 1                                                                                                                            | 0.1                |
| Wave        | acid.wave        | 0 .. 1                                                                                                                             | 0                  |
| Pulse Width | acid.pulse_width | 0.05 .. 0.95                                                                                                                       | 0.5                |
| Sub Osc     | acid.sub_osc     | 0 .. 1                                                                                                                             | 0                  |
| Filter      | acid.filter      | Moog Ladder, Prophet (SSM2040), Juno (IR3109), Oberheim SEM, Xpander, Diode Ladder (303), Korg35 (MS-20), Polivoks, EDP Wasp, Comb | Diode Ladder (303) |
| Cutoff      | acid.cutoff      | 40 .. 12000 Hz                                                                                                                     | 600 Hz             |
| Resonance   | acid.resonance   | 0 .. 1                                                                                                                             | 0.72               |
| Env Amount  | acid.env_amount  | 0 .. 6 oct                                                                                                                         | 2.5 oct            |
| Decay       | acid.decay       | 5 .. 3000 ms                                                                                                                       | 220 ms             |
| Accent      | acid.accent      | 0 .. 1                                                                                                                             | 0.6                |
| Attack      | acid.amp_attack  | 0.5 .. 500 ms                                                                                                                      | 3 ms               |
| Amp Decay   | acid.amp_decay   | 5 .. 3000 ms                                                                                                                       | 250 ms             |
| Sustain     | acid.amp_sustain | 0 .. 1                                                                                                                             | 0.8                |
| Release     | acid.amp_release | 5 .. 3000 ms                                                                                                                       | 60 ms              |
| Glide       | acid.glide       | 0 .. 500 ms                                                                                                                        | 60 ms              |
| Drive       | acid.drive       | 0 .. 1                                                                                                                             | 0.5                |
| Key Track   | acid.key_track   | 0 .. 1                                                                                                                             | 0.3                |
| Low Cut     | acid.low_cut     | 20 .. 500 Hz                                                                                                                       | 120 Hz             |
| High Cut    | acid.high_cut    | 500 .. 20000 Hz                                                                                                                    | 9000 Hz            |

|              |                   |              |        |
|--------------|-------------------|--------------|--------|
| Dub Send     | acid.dub_send     | 0 .. 1       | 0.25   |
| Room Send    | acid.room_send    | 0 .. 1       | 0.01   |
| Duck         | acid.duck         | 0 .. 24 dB   | 4 dB   |
| Duck Release | acid.duck_release | 30 .. 600 ms | 200 ms |

## CHORD

| Control     | Key               | Range                | Default   |
|-------------|-------------------|----------------------|-----------|
| Level       | chord.level       | -60 .. 6 dB          | -5 dB     |
| Width       | chord.width       | 0 .. 1               | 0.6       |
| Detune      | chord.detune      | 0 .. 30 ct           | 10 ct     |
| Octave      | chord.octave      | -2 .. 2              | 0         |
| Attack      | chord.attack      | 0.5 .. 500 ms        | 10 ms     |
| Decay       | chord.decay       | 20 .. 3000 ms        | 300 ms    |
| Sustain     | chord.sustain     | 0 .. 1               | 0.15      |
| Release     | chord.release     | 10 .. 3000 ms        | 400 ms    |
| Brightness  | chord.bright      | 200 .. 12000 Hz      | 2500 Hz   |
| Env Amount  | chord.env_amount  | 0 .. 5 oct           | 1.5 oct   |
| Band        | chord.band        | 100 .. 3000 Hz       | 380 Hz    |
| Band Mode   | chord.band_mode   | Band Pass, High Pass | Band Pass |
| Band Q      | chord.band_q      | 0.2 .. 4             | 0.7       |
| Sweep       | chord.sweep       | 0 .. 2 oct           | 0.5 oct   |
| Sweep Rate  | chord.sweep_rate  | 0.01 .. 2 Hz         | 0.2 Hz    |
| Bits        | chord.crush       | 4 .. 16              | 8         |
| Crush Mix   | chord.crush_mix   | 0 .. 1               | 0.2       |
| Phaser      | chord.phaser      | 0 .. 1               | 0.5       |
| Phaser Rate | chord.phaser_rate | 0.05 .. 4 Hz         | 0.5 Hz    |
| Dip 600 Hz  | chord.dip         | -18 .. 0 dB          | -7 dB     |
| Dub Send    | chord.dub_send    | 0 .. 1               | 0.5       |
| Plate Send  | chord.plate_send  | 0 .. 1               | 0.35      |

## DRONE

| Control    | Key              | Range         | Default |
|------------|------------------|---------------|---------|
| Level      | drone.level      | -60 .. 6 dB   | -13 dB  |
| Octave     | drone.octave     | -2 .. 2       | 0       |
| Detune     | drone.detune     | 0 .. 30 ct    | 8 ct    |
| Cutoff     | drone.cutoff     | 80 .. 8000 Hz | 700 Hz  |
| Resonance  | drone.resonance  | 0 .. 1        | 0.25    |
| Sweep      | drone.sweep      | 0 .. 4 oct    | 1.5 oct |
| Sweep Bars | drone.sweep_bars | 4 .. 128 bars | 32 bars |

|            |                  |              |      |
|------------|------------------|--------------|------|
| Attack     | drone.attack     | 0.01 .. 20 s | 4 s  |
| Release    | drone.release    | 0.01 .. 20 s | 6 s  |
| Plate Send | drone.plate_send | 0 .. 1       | 0.4  |
| Room Send  | drone.room_send  | 0 .. 1       | 0.02 |

**TEXTURE**

| Control  | Key             | Range       | Default |
|----------|-----------------|-------------|---------|
| Level    | texture.level   | -60 .. 0 dB | -26 dB  |
| Crackle  | texture.crackle | 0 .. 1      | 0.5     |
| Hum      | texture.hum     | 0 .. 1      | 0.2     |
| Hum Freq | texture.hum_hz  | 40 .. 70 Hz | 50 Hz   |
| Erosion  | texture.erosion | 0 .. 1      | 0.4     |
| Width    | texture.width   | 0 .. 1      | 0.8     |

# Dub



The dub chain: a tape echo with wow, flutter and drive, a spring, a plate. The ping, the chord and the kit's buses send into it, and the composer throws single hits into the echo on the 8-bar lines. The room is the hall the hats, the percussion and the ping live in; the grain cloud hears the ping and the chord and scatters them.

## DUB

| Control       | Key              | Range                          | Default |
|---------------|------------------|--------------------------------|---------|
| Echo Time     | dub.echo_time    | 1/16, 1/8, 3/16, 1/4, 3/8, 1/2 | 3/16    |
| Feedback      | dub.feedback     | 0 .. 1.2                       | 0.25    |
| Echo Tone     | dub.tone         | 800 .. 12000 Hz                | 4500 Hz |
| Echo Low Cut  | dub.low_cut      | 40 .. 800 Hz                   | 200 Hz  |
| Wow           | dub.wow          | 0 .. 3 ms                      | 0.5 ms  |
| Flutter       | dub.flutter      | 0 .. 0.5 ms                    | 0.05 ms |
| Echo Drive    | dub.drive        | 0 .. 18 dB                     | 4 dB    |
| Echo Return   | dub.echo_return  | -60 .. 6 dB                    | -3 dB   |
| Spring Return | dub.spring       | -60 .. 6 dB                    | -9 dB   |
| Spring Decay  | dub.spring_decay | 0.3 .. 6 s                     | 2 s     |

|                 |                    |              |        |
|-----------------|--------------------|--------------|--------|
| Plate Decay     | dub.plate_decay    | 0.5 .. 12 s  | 4.2 s  |
| Plate Damping   | dub.plate_damping  | 0 .. 1       | 0.4    |
| Plate Pre-Delay | dub.plate_predelay | 0 .. 100 ms  | 20 ms  |
| Plate Low Cut   | dub.plate_low_cut  | 60 .. 800 Hz | 250 Hz |
| Plate Return    | dub.plate_return   | -60 .. 18 dB | 6 dB   |
| Ping to Echo    | dub.ping_send      | 0 .. 1       | 0.12   |
| Hats to Echo    | dub.hats_send      | 0 .. 1       | 0      |
| Perc to Echo    | dub.perc_send      | 0 .. 1       | 0      |

## SPACE

| Control   | Key             | Range            | Default |
|-----------|-----------------|------------------|---------|
| Return    | space.level     | -60 .. 12 dB     | 0 dB    |
| Size      | space.size      | 0.3 .. 3         | 0.9     |
| Decay     | space.decay     | 0.2 .. 12 s      | 1.2 s   |
| Damping   | space.damping   | 0 .. 1           | 0.5     |
| Pre-Delay | space.predelay  | 0 .. 100 ms      | 15 ms   |
| Low Cut   | space.low_cut   | 60 .. 800 Hz     | 250 Hz  |
| High Cut  | space.high_cut  | 1000 .. 18000 Hz | 6000 Hz |
| Hats Send | space.hats_send | 0 .. 1           | 0.25    |
| Perc Send | space.perc_send | 0 .. 1           | 0.3     |
| Ping Send | space.ping_send | 0 .. 1           | 0.4     |

## CLOUD

| Control    | Key              | Range         | Default |
|------------|------------------|---------------|---------|
| Level      | cloud.level      | -60 .. 6 dB   | -60 dB  |
| Density    | cloud.density    | 0.5 .. 60 /s  | 12 /s   |
| Grain Size | cloud.size       | 20 .. 1000 ms | 180 ms  |
| Pitch      | cloud.pitch      | 0 .. 1        | 0.3     |
| Spray      | cloud.spray      | 0.05 .. 4.5 s | 2 s     |
| From Ping  | cloud.ping_send  | 0 .. 1        | 0.8     |
| From Chord | cloud.chord_send | 0 .. 1        | 0.6     |
| Plate Send | cloud.plate_send | 0 .. 1        | 0.5     |

# Mixer and Master



The meters: each deck after its mixer channel, the output, and the loudness of the last 400 ms. Each deck's track bus has a group high pass (the form's filter sweeps), a tilt of the highs against the lows, and the glue -- a bus compressor mixed in at 35 per cent -- after which the loudness correction sits. The master keeps the side under Mono Below mono, sets the level, has an optional vinyl cut, a clipper at four times the rate and a true-peak limiter.

The motion is the free-running modulation of the whole instrument: the hats' cutoff and level, their decay, the percussion's cutoff, the rumble's drive, slow LFOs of prime periods that never line up.

A deck's channel is a DJ mixer's: the fader, a three-band isolator (200 Hz and 2.5 kHz, Linkwitz-Riley, each band from a kill to +6 dB), a filter that is a low pass to the left and a high pass to the right, and the send into the mixer's effects. In a set the composer moves them as a DJ would; in the plugin they are yours too.

## MIX

| Control         | Key                         | Range           | Default  |
|-----------------|-----------------------------|-----------------|----------|
| Hats Level      | <code>mix.hats_level</code> | -24 .. 12 dB    | -3.5 dB  |
| Hats Cut        | <code>mix.hats_cut</code>   | 200 .. 20000 Hz | 20000 Hz |
| Perc Level      | <code>mix.perc_level</code> | -24 .. 12 dB    | -6.5 dB  |
| Perc Cut        | <code>mix.perc_cut</code>   | 200 .. 20000 Hz | 20000 Hz |
| Drum Saturation | <code>mix.drum_sat</code>   | 0 .. 1          | 0.35     |
| Duck Low        | <code>mix.duck_low</code>   | 0 .. 18 dB      | 10 dB    |
| Duck Mid        | <code>mix.duck_mid</code>   | 0 .. 12 dB      | 3 dB     |
| Group Low Cut   | <code>mix.low_cut</code>    | 20 .. 500 Hz    | 20 Hz    |

## MASTER

| Control        | Key                            | Range        | Default |
|----------------|--------------------------------|--------------|---------|
| Level          | <code>master.level</code>      | -24 .. 12 dB | 0 dB    |
| Glue Threshold | <code>master.threshold</code>  | -40 .. 0 dB  | -16 dB  |
| Glue Ratio     | <code>master.ratio</code>      | 1 .. 10      | 2       |
| Clip Drive     | <code>master.clip</code>       | 0 .. 9 dB    | 2 dB    |
| Ceiling        | <code>master.ceiling</code>    | -6 .. 0 dBTP | -1 dBTP |
| Mono Below     | <code>master.mono_below</code> | 40 .. 250 Hz | 120 Hz  |
| Tilt           | <code>master.tilt</code>       | -12 .. 12 dB | 10 dB   |
| Vinyl Cut      | <code>master.cut</code>        | off / on     | off     |

## MOTION

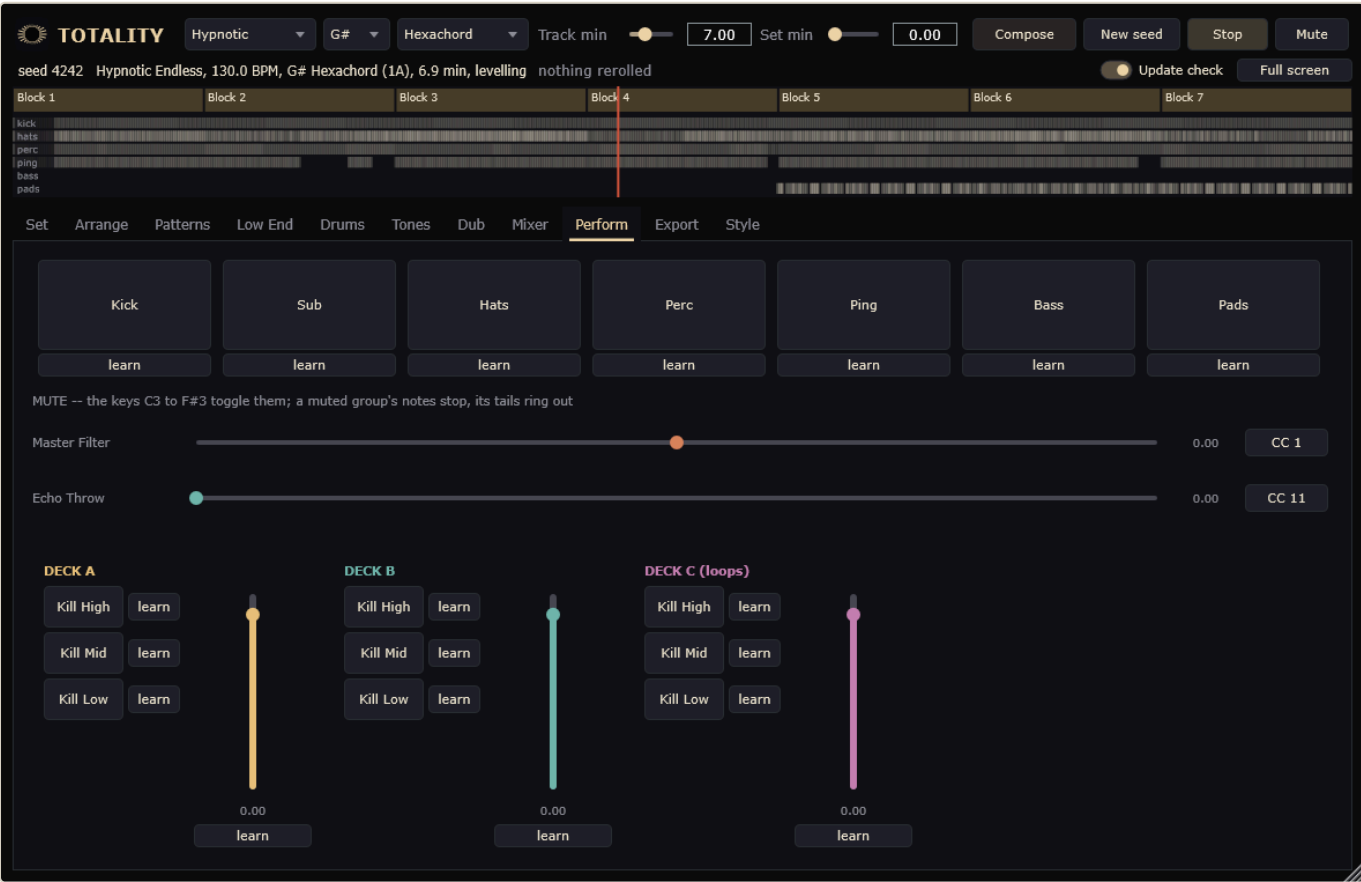
| Control      | Key                              | Range        | Default |
|--------------|----------------------------------|--------------|---------|
| Amount       | <code>motion.amount</code>       | 0 .. 1       | 1       |
| Hats Cut     | <code>motion.hats_cut</code>     | 0 .. 1.5 oct | 0.4 oct |
| Hats Level   | <code>motion.hats_level</code>   | 0 .. 6 dB    | 1.5 dB  |
| Hat Decay    | <code>motion.hat_decay</code>    | 0 .. 0.5     | 0.2     |
| Perc Cut     | <code>motion.perc_cut</code>     | 0 .. 1.5 oct | 0.4 oct |
| Rumble Drive | <code>motion.rumble_drive</code> | 0 .. 6 dB    | 2 dB    |

## DECK (3 INSTANCES)

| Control | Key                      | Range       | Default |
|---------|--------------------------|-------------|---------|
| Fader   | <code>deckN.fader</code> | -60 .. 6 dB | 0 dB    |
| Low     | <code>deckN.low</code>   | -60 .. 6 dB | 0 dB    |
| Mid     | <code>deckN.mid</code>   | -60 .. 6 dB | 0 dB    |

|         |               |             |      |
|---------|---------------|-------------|------|
| High    | deckN.high    | -60 .. 6 dB | 0 dB |
| Filter  | deckN.filter  | -1 .. 1     | 0    |
| FX Send | deckN.fx_send | 0 .. 1      | 0    |

# Perform



As at a mixer. Seven mutes: a muted group's notes stop and its tails ring out -- the kick out, the classic move. The master filter takes the whole mix, the echo throw sends it into the mixer's tape echo. Each deck has its kills and its fader. Every control can be learned from a MIDI controller: click learn, move the controller (a right click forgets it). By default the mod wheel moves the filter and the expression pedal the throw, and the keys from middle C up (C to F#) toggle the mutes of kick, sub, hats, perc, ping, bass and pads. The bindings are saved with the project.

The perform controls are live only: an export or tot\_render plays the track as it was composed.

## PERFORM

| Control       | Key               | Range    | Default |
|---------------|-------------------|----------|---------|
| Master Filter | perform.filter    | -1 .. 1  | 0       |
| Echo Throw    | perform.throw     | 0 .. 1   | 0       |
| Mute Kick     | perform.mute_kick | off / on | off     |
| Mute Sub      | perform.mute_sub  | off / on | off     |
| Mute Hats     | perform.mute_hats | off / on | off     |
| Mute Perc     | perform.mute_perc | off / on | off     |
| Mute Ping     | perform.mute_ping | off / on | off     |
| Mute Bass     | perform.mute_bass | off / on | off     |
| Mute Pads     | perform.mute_pads | off / on | off     |

# Export and cues

TOTALITY

Hypnotic

G#

Hexachord

Track min

7.00

Set min

0.00

Compose

New seed

Stop

Mute

seed 4242 Hypnotic Endless, 130.0 BPM, G# Hexachord (1A), 6.9 min, levelling nothing rerolled

Block 1

Block 2

Block 3

Block 4

Block 5

Block 6

Block 7

kick

hats

perc

ping

bass

pads

Set

Arrange

Patterns

Low End

Drums

Tones

Dub

Mixer

Perform

Export

Style

WAV + MIDI + cues

... with stems

... with DJ loops

... with both

Save .totset

Load .totset

seed 4242 Hypnotic Endless, 130.0 BPM, G# Hexachord (1A), 6.9 min, levelling

The export renders what plays as it was composed (the performer's mutes, filter and throw are live only) at 48 kHz:

- a 24-bit WAV with its cues as markers (the bass's entry, the kick-outs and returns, the outro; in a set every track, swap and loop) and the same cues as JSON beside it (<name>.wav.cues.json),

- the MIDI file with the tempo map and a channel per part,

- with stems: a 32-bit WAV per element into <name>\_stems; their sum is the mix before the master, exactly,

- with DJ loops (a track): 4 and 8 bars of its loudest block, seamless, the mix and kick, hats and perc alone.

An .totset holds the seed, the lengths, the rerolls and every changed knob.

OSC CUES

OSC Cues

OSC Port

9000

TOTALITY

Hypnotic

A

Aeolian

Track min

7.00

Set min

40.0

Compose

New seed

Stop

Mute

seed 5 set of 9 tracks, Peak, 43.0 min, levelling nothing rerolled

T1 Hypnotic 8A

T2 Hypnotic 8A

T3 Hypnotic 8A

T4 Ostgut 9A

T5 Ostgut 8A

T6 Ostgut 9A

T7 Ostgut 8A

T8 Ostgut 7A

T9 Ostgut 8A

kick

hats

perc

ping

bass

pads

Set

Arrange

Patterns

Low End

Drums

Tones

Dub

Mixer

Perform

Export

Style

Track 3 of 9: Hypnotic, Arc, 129.5 BPM, A Aeolian (8A), 224 bars, the sub owns the low end, bar similarity 0.95; figure: the bass from bar 33, waves land on 97 113 145 193

Sounds: kick Sunken Forge, rumble Misty Grind, sub Round Tone, ping Shimmering Chirp, bass High Wreath, acid Laser Tamarind, chord Gleaming Mash, drone Quiet Diapason, texture Open Rustle

reroll form

reroll harmony

reroll rack

reroll layers

reroll blocks

reroll events

reroll hands

reroll sounds

reroll figure

reroll the whole track

reroll the set's plan

T1 Hypnotic 8A

T2 Hypnotic 8A

T3 Hypnotic 8A

T4 Ostgut 9A

T5 Ostgut 8A

T6 Ostgut 9A

T7 Ostgut 8A

T8 Ostgut 7A

T9 Ostgut 8A

kick

hats

perc

ping

bass

pads

A set of 40 minutes on the Arrange page: nine tracks on two decks, their blends, the swaps.

The export renders what plays at 48 kHz: a 24-bit WAV with cue markers (the bass's entry, the kick-outs and returns, the outro; in a set every track, swap and loop) and the same cues as JSON beside it, for DJ software that does not read the WAV's markers; the MIDI file with the tempo map and a channel per part; on request the stems -- a WAV per element, whose sum is exactly the mix before the master -- and the DJ loops of a track: 4 and 8 bars of its loudest block, seamless, as the mix and as kick, hats and percussion alone. An .totset file holds the seed, the lengths, the rerolls and every changed knob.

The OSC cues tell a visualiser such as Kaleidoscope what the score knows, at the moment it is heard: the beat and the tempo, the bar, the block's name, the form's operation, the key. They go to this machine at the port given, or to TOT\_CUE\_HOST.

```
/tot/beat i f      beat number, tempo
/tot/bar i        bar number
/tot/block s      the block's name ("Intro", "Block 3", "Reduction", a set's track)
/tot/op s         the operation ("add ride", "kick out")
/tot/key s        the key as a Camelot label ("8A")
```

CUE

| Control  | Key         | Range         | Default |
|----------|-------------|---------------|---------|
| OSC Cues | cue.enabled | off / on      | off     |
| OSC Port | cue.port    | 1024 .. 65535 | 9000    |

# Style



The numbers of the profile the knobs describe, and the references' measurements per style that the profiles are fitted to. A style of your own: Take the profile's numbers, change what you like, switch Use on -- the tempo range, the weights of the three forms, the length, the rack's mutation and reroll, polymeters, fills, swing, the filtered edges, the longest kick-out, the events, the corridor of bar similarity the composer's candidates are chosen against, and the loudness the Leveler brings each track to.

## CUSTOM

| Control          | Key                | Range          | Default |
|------------------|--------------------|----------------|---------|
| Use Custom Style | custom.use         | off / on       | off     |
| Tempo Low        | custom.bpm_low     | 120 .. 140 BPM | 128 BPM |
| Tempo High       | custom.bpm_high    | 120 .. 140 BPM | 133 BPM |
| Arc Weight       | custom.arc         | 0 .. 1         | 0.5     |
| Peak Weight      | custom.peak        | 0 .. 1         | 0.2     |
| Endless Weight   | custom.endless     | 0 .. 1         | 0.3     |
| Sub Owns Low     | custom.sub_chance  | 0 .. 1         | 0.3     |
| Blocks Low       | custom.blocks_low  | 4 .. 12        | 6       |
| Blocks High      | custom.blocks_high | 4 .. 12        | 8       |
| Mutation         | custom.mutation    | 0 .. 1         | 0.25    |
| Reroll           | custom.reroll      | 0 .. 1         | 0.15    |

|                  |                      |                |          |
|------------------|----------------------|----------------|----------|
| Polymer          | custom.polymer       | 0 .. 1         | 0.8      |
| Fills            | custom.fill          | 0 .. 1         | 0.15     |
| Filtered Edges   | custom.edge          | 0 .. 1         | 0.5      |
| Longest Kick-out | custom.max_reduction | 4 .. 32 bars   | 32 bars  |
| Swing Low        | custom.swing_low     | 50 .. 62 %     | 51 %     |
| Swing High       | custom.swing_high    | 50 .. 62 %     | 56 %     |
| Events           | custom.event_rate    | 0 .. 1         | 0.35     |
| Throw Share      | custom.throw_share   | 0 .. 1         | 0.5      |
| Density Cap      | custom.density_cap   | 3 .. 12        | 9        |
| Bar Similarity   | custom.similarity    | 0.7 .. 1       | 0.9      |
| Target LUFS      | custom.peak_lufs     | -14 .. -7 LUFS | -10 LUFS |

## Meta Quest

---

The whole generator runs on the headset (build-quest\TotalityQuest.apk, see Quest/README.md). The hands are the hands at the mixer: left pinch play and stop, right pinch kick out and in, both hands together the next track; the left hand's height the master filter, the right's the echo throw. Above and ahead of the player the Eclipse turns at the size of a room. A file tot.cfg in the app's folder sets the seed, the style, a set's length and any knob.

## The command line

---

tot\_render composes and renders without a window, the same samples the plugin plays:

```
tot_render --seed 7 --out track.wav --midi track.mid --stems stems --loops loops
tot_render --seed 2026 --set-minutes 120 --out set.wav --score-json set.json
tot_render --set "compose.style=Dub; compose.minutes=8" --seed 3 --out dub.wav
tot_render --list                every parameter with its value
tot_render --bench --quality quest how fast, at the headset's quality
```

The evaluation (Tools/eval\_report.py) measures rendered tracks as the references were measured and writes a report of what lies outside their corridor.